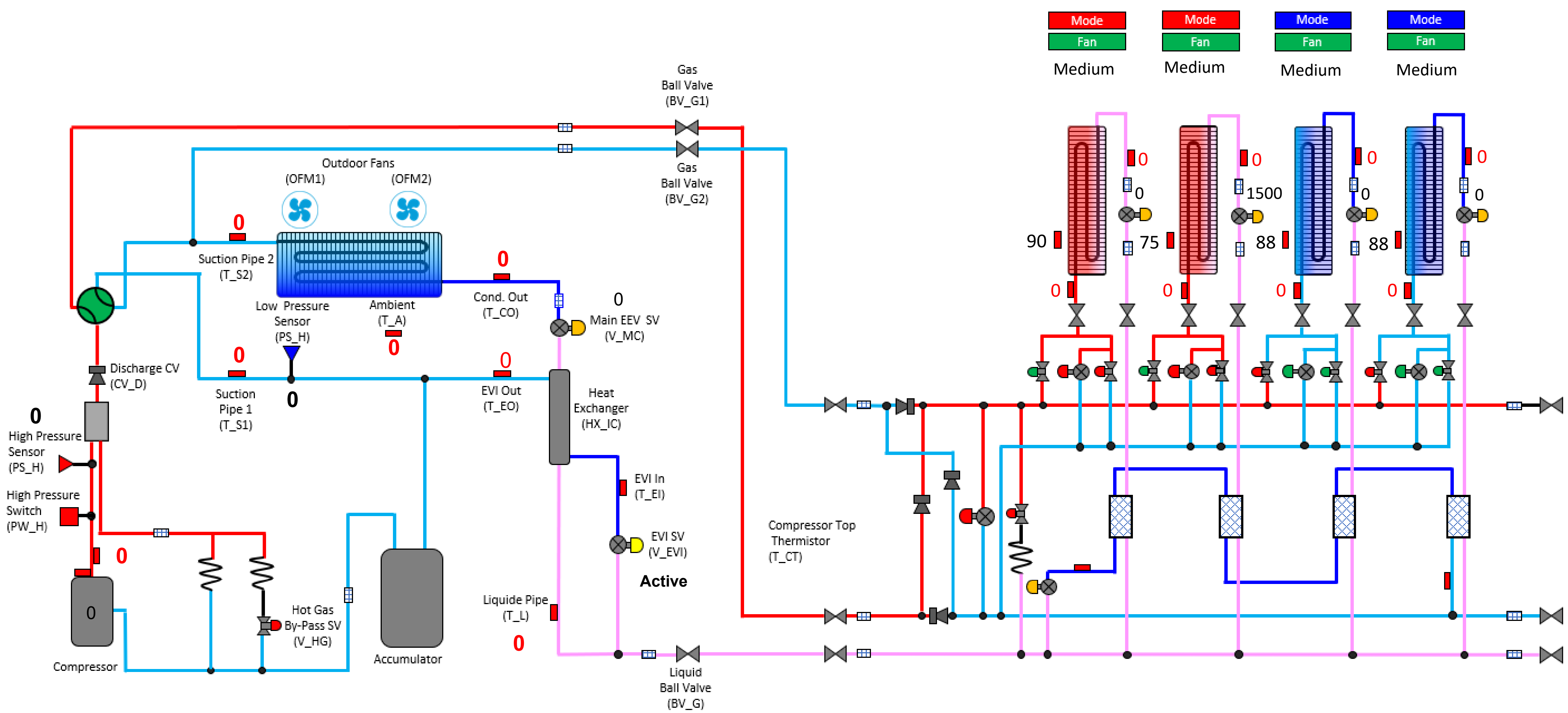
DVM S Eco Heat Recovery Operation Evaluation
Heating Main Mode



Comments:

Outdoor Unit			
Component	**Target**	**Reading**	
Ambient Temp.		0	
Compressor Hz.	20-160	0	Ok
High pressure	427 psig	0	Ok
H.P Sat. Temp	121.5^0		
Discharge Temp.	150^0-212^0	0	Ok
EVI Out	Below 194^0	0	Ok
Low pressure		0	Ok
L.P. Sat. Temp	OAT - 2^0 - 10^0		
Suction 1 Temp.		0	Ok
Suction 2 Temp		0	Ok
S.H.	3.6^0 or 9^0	***0***	Ok
Main EEV Position		0	Ok
Cond. Out Temp.	OAT - 2^0 -10^0	0	Ok
Liquid Line Temp.		0	Ok
Sub-Cooling @ T_L	9^0 +	***0***	Ok

Indoor Units			
Component	**Target**	**Reading**	
IDU 1 EEV	500 – 1500 Steps	0	Ok
IDU 1 SC	9^0	***0***	Ok
IDU 1 T PI		0	Ok
IDU 1 T PO		0	Ok
IDU 2 EEV	500 – 1500 Steps	1500	Ok
IDU 2 SC	9^0	***0***	Ok
IDU 2 T PI		0	Ok
IDU 2 T PO		0	Ok
IDU 3 EEV	250 – 1400 Steps	0	Ok
IDU 3 SH	3.6^0 (Typical 0^0 -7^0)	***0***	Ok
IDU 3 T PI		0	Ok
IDU 3 T PO		0	Ok
IDU 4 EEV	250 – 1400 Steps	0	Ok
IDU 4 SH	3.6^0 (Typical 0^0 -7^0)	***0***	Ok
IDU 4 T PI		0	Ok
IDU 4 T PO		0	Ok

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature and EEV steps are entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

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DVM S Heat Pump Operation Evaluation Heating Main Mode

Air Source Outdoor Unit

Component	Abbreviation		Heating Main Mode	Comments
Compressor	Comp	Operation	Target High pressure control (default: 427 psig)	Compressors primary function is to maintain head pressure
		Range	20~160Hz	
Outdoor Fan Motor	OFM	Operation	Low pressure control	Low Side saturation temperature is "typically" 2-10 degrees lower than outside air temperature
		Range	0-39 step (0-1300rpm)	
Main EEV	E_M	Operation	*EVI Superheat control (default : SH 3.6°F or 9°F)	Modulates to maintain 3.6 or 9 degrees of evaporator superheat
		Range	125 - 2000 Steps	
EVI EEV	E_EVI	Operation	On (Open)	Discharge protecion or Flash-vapor injection
		Range	0-480step	
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 59°F or all compressor in unit are at 50Hz↓	
			Off (Close): T_a < 50°F and all compresors in unit are at 70Hz↑	Low Ambient heating Operation and discharge temperature protection. Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compresor is running)	**Off** during start up if DSH is < 18°F. **Off** for 2 min. if DSH is < 18°F **On** when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	Off (Closed)	
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the outside air temperature	Typicaly it ranges between 2 to 10 degrees below outside air temperature

Mode Control Unit (MCU)

Component	Abbreviation		Heating Main Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open) Call for Heating Off (Closed) Call for Cooling	
Port Cooling Solenoid	Cool	Operation	On (Open) Call for Cooling Off (Closed) Call for Heat	
Port EEV	EEV	Operation	Closed 0 Steps	Open when a zone satisfies or for mode change
		Range	0 - 480 Steps	
Sub-Cooling EEV	SC_EEV	Operation	Active in heating Main	Provide subcooling for cooling zones
		Range	Modulating	Maintains 5.4 degrees of superheat between pipe in and out temps.
Sub-Cooler In Temperature		Operation	Should be colder then Pipe Out	
Sub-Cooler Out Temperature		Operation	Should be hotter then pipe in.	

Indoor Unit IDU

Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed:|Ts - Ti| ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < |Ts - Ti| ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < |Ts - Ti|	

Indoor Unit IDU Cooling Mode

Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp (Typical range 0-7 degrees)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1400 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed:|Ts - Ti| ≤ 1.8°F	TS - Set Temperature
			Medium speed: 1.8°F < |Ts - Ti| ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < |Ts - Ti|	

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